

Exploring the Design Space of Adaptive Explanations in Automated Vehicles

ANONYMOUS AUTHOR(S)



Fig. 1. The simulator platform combines a motion-based, VR-enabled automated-driving environment, scripted traffic incidents, an in-vehicle explanation interface, and a feedback loop able to adapt explanation behavior through repeated exposure.

Explanations in automated vehicles have been proposed to support users' understanding, trust, and acceptance of automated driving behavior. However, many explanation interfaces are still designed as static interventions, even though users' information needs may vary depending on driving context, familiarity, perceived criticality, and personal preference. This work presents a simulator platform for prototyping adaptive in-vehicle explanations in automated driving. The platform combines a Unity-based driving environment, a motion-based, VR-enabled simulator setup, scripted traffic incidents, an in-vehicle explanation interface, and a feedback loop to adapt future explanations. The system uses a large language model-based explanation pipeline that receives structured scenario information and user feedback to determine whether an explanation should be presented, how detailed it should be, and how it should be delivered. We describe the platform, outline the design space it enables, and reflect on how such a system can support future AutomotiveUI research on adaptive explanations, explanation suppression, user control, and human-AI interaction in automated vehicles.

CCS Concepts: • **Human-centered computing** → *User studies*.

Additional Key Words and Phrases: Automated vehicles, Automotive user interfaces, In-vehicle explanations, Adaptive interfaces, Explainable AI, Large language models, Driving simulation, Human-AI interaction

ACM Reference Format:

Anonymous Author(s). 2026. Exploring the Design Space of Adaptive Explanations in Automated Vehicles. In *18th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI '26)*, September 20–23, 2026, Gothenburg, Sweden. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3640792.3675722>

1 Introduction

Automated vehicles increasingly require users to interpret complex system behavior in situations where control, responsibility, and understanding are distributed between human and machine. As vehicles become more capable, users may no longer directly control every maneuver, but they still need to understand why the vehicle behaves in certain

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

Manuscript submitted to ACM

Manuscript submitted to ACM

ways. This is especially relevant in ambiguous or safety-critical situations, such as sudden braking, evasive maneuvers, lane changes, or interactions with vulnerable road users.

Prior work in AutomotiveUI and Explainable AI (XAI) has investigated in-vehicle explanations as a way to support transparency, user understanding, trust calibration, and acceptance of automated vehicle behavior [2, 3, 8, 10, 12]. Explanations may tell users why the vehicle acted, what it perceived, what it intends to do next, or how confident it is in a situation. However, explanation needs are unlikely to be static. Information that is useful during early encounters with automation may become unnecessary or annoying after repeated exposure. Conversely, users may still want explanations in unexpected, ambiguous, or safety-relevant situations.

This creates a design challenge for future automated vehicles: they may need to decide not only *what* to explain, but also *when* to explain, *how much* to explain, *how* to explain, and *when not to explain at all*. This shifts explanation design from static content design toward adaptive interaction design. Instead of asking only which explanation format is best, researchers and designers must also ask how explanation behavior should change across situations, users, and time.

To support this exploration, we present a simulator platform for prototyping adaptive in-vehicle explanations in automated driving. The platform combines a Unity-based driving simulation, a digital twin environment, scripted traffic incidents, VR-based passenger experience, audio and text explanation delivery, and an in-simulator feedback interface. The system allows researchers to author scenarios, manipulate contextual properties, present explanations, collect user preferences, and use these preferences to adapt future explanation behavior.

As an exploratory implementation, we integrated a large language model-based explanation pipeline. The pipeline receives structured information about the current driving scenario, the incident encountered, explanation constraints, and stored user feedback. It then generates explanation content and determines whether the explanation should be delivered or suppressed. We use this implementation not as a claim that unconstrained generative models should be deployed in production vehicles, but as a flexible prototyping mechanism for exploring adaptive explanation behavior. This Work-in-Progress contributes:

- **A simulator platform for adaptive in-vehicle explanations.** We present a VR-enabled and motion-based automated-driving simulator that allows researchers to prototype explanation behavior in scripted traffic situations.
- **An adaptive explanation pipeline.** We describe how structured scenario data and explicit user feedback can be used to adapt explanation frequency, detail, modality, and tone.
- **A design space for adaptive explanations.** We outline design dimensions including timing, explanation absence, modality, personalization source, adaptation target, and safety constraints. We also discuss how the platform can support future research on explanation suppression, user control, context-sensitive explanations, and LLM-supported prototyping.

2 Background and Motivation

XAI seeks to enhance the transparency of AI decisions by enabling human users to better understand the system's behavior through explaining the decision rationale [4]. Prior research emphasizes that XAI can also contribute to better trust calibration and higher perceived safety in the context of autonomous driving [6, 9, 12, 13]. The proposals in the literature for explaining vehicle behavior range from visual cues (e.g., trajectory previews of the car or detected pedestrians [2, 3] or upcoming maneuvers [12]) over auditory feedback (e.g. in form of sounds [7]) to natural language

explanations incorporating anthropomorphism [5] or even avatar-like HMIs to increase users' understanding [1]. In addition to user-related factors, the requirement for explanations is strongly shaped by the situational context [13].

Despite these promising advances in the use of XAI in cars, some challenges remain. Findings by Sadeghi et al. [11] highlight that the benefits of explanations in complex scenarios must be considered carefully. A one-size-fits-all explanation approach cannot accommodate the diverse and changing needs of vehicle users, as excessive information may distract or annoy, while insufficient information may cause confusion. Adaptive systems should therefore consider both context and user preferences, providing detailed explanations for unfamiliar, unexpected, critical, or ambiguous events while suppressing explanations that become unnecessary through repeated exposure. Driving simulators offer a safe, controlled environment for studying these dynamics through repeatable scenarios, contextual manipulation, interface prototyping, repeated interaction, and user feedback. Our platform combines simulated incidents, generated explanations, and user feedback to test and prototype when automated vehicles should explain, what they should explain, or remain silent.

3 Simulator Platform

The simulator platform consists of four main components: a virtual driving environment, scripted scenario authoring, an in-vehicle explanation interface, and an adaptive explanation pipeline.

3.1 Driving Environment and Scenario Authoring

The driving environment is implemented in Unity 3D and represents automated driving in a digital twin of [BLANKED], combining satellite and OpenStreetMap data to create realistic urban and highway settings. The simulator includes a motion base with two Next Level Motion Plus platforms for roll and pitch and uses a VR headset to provide an immersive experience from inside the automated vehicle. Although steering-wheel and pedal input are supported, the current platform focuses on the passenger experience in a fully automated vehicle. Users observe rather than control the vehicle, allowing researchers to study how future passengers or fallback users interpret and respond to automated-driving behavior.

The platform supports scripted scenarios involving sudden braking, blocked roads, lane changes, intersection conflicts, unexpected behavior by other road users, and changes in traffic density. Each incident can be annotated with contextual properties relevant to explanation design, currently including:

- **Incident criticality:** whether the situation is low or high in perceived safety relevance.
- **Traffic density:** whether the surrounding traffic is sparse or dense.
- **Road type:** whether the incident occurs in an urban or highway environment.
- **Incident type:** the kind of event, such as braking, lane change, obstacle avoidance, or interaction with another road user.
- **Incident description:** a structured natural language description of what happened and why the automated vehicle reacted.
- **Further Details:** Further information about the incident if the user asks for more details.

This scenario structure allows researchers to manipulate driving context while keeping the vehicle behavior repeatable. It also allows the explanation pipeline to receive structured information about the current situation.

3.2 Scenario Context Model

A central goal of the platform is to make driving context available to the explanation system in a structured way. Instead of generating explanations from raw simulation data alone, each scenario contains a context model that describes the event in terms relevant to explanation design. For example, a scripted incident may be represented as follows:

```
{
  "roadType": "urban",
  "trafficDensity": "high",
  "criticality": "high",
  "incidentType": "sudden_braking",
  "description": "The vehicle braked because another car entered the intersection unexpectedly.",
  "furtherDetails": "Another road user pulled out of the right street, despite having a stop sign",
  "vehicleAction": "braked",
  "visibleCause": "vehicle from the left entered the intersection",
  "explanationConstraints": {
    "maxSentences": 3,
    "avoidJargon": true,
    "stayFactual": true
  }
}
```

This representation separates the driving event from the explanation. The same incident can therefore be explained in different ways depending on user preference, modality, desired detail, or system policy. For instance, one version may provide a concise verbal explanation, while another may show a short text notification or suppress the explanation entirely.

The context model is intentionally simple at this stage. It is not intended to fully represent the complexity of real-world automated driving. Instead, it provides a controllable abstraction that supports early-stage interface prototyping and systematic scenario comparison.

3.3 In-Vehicle Explanation Interface

The in-vehicle interface delivers explanations as text on the center console and through text-to-speech, supporting both glanceable visual information and auditory communication. Explanations can vary in intrusiveness, audio is more immediate and attention-demanding, text is more passive and reviewable, and a minimal notification can indicate when a full explanation is omitted or unnecessary. After each incident, users can provide feedback on how the system should handle similar situations, using three dimensions:

- **Explanation frequency:** whether explanations in similar situations should be kept, reduced, or omitted.
- **Level of detail:** whether future explanations should contain more detail, the same amount of detail, or less detail.
- **Tone:** whether future explanations should be more reassuring, remain neutral, or be less emotionally expressive.

This feedback mechanism makes preference collection part of the interaction loop. Rather than treating explanation preferences as static settings, the system can collect situated feedback after specific events and use it for later adaptation.

3.4 Adaptive Explanation Pipeline

The adaptive explanation pipeline uses scenario context and user feedback to determine explanation behavior. In the current implementation, a large language model is used as a flexible prototyping component. The model receives structured scenario information, explanation constraints, and available preference history. It then returns a structured response that can be processed by the simulator.

The response contains:

- **Explanation text:** the content that should be shown or spoken to the user.
- **Delivery mode:** whether the explanation should be delivered as audio, text, both, or suppressed.
- **Style instruction:** guidance for text-to-speech delivery, such as concise, calm, or reassuring.
- **Reasoning trace:** a short explanation of why the system selected this explanation behavior.

The reasoning trace is not presented to the user by default. Instead, it is logged for researchers and designers. This allows the design team to inspect why the system decided to deliver, alter, or suppress an explanation. Such inspectability is important because adaptive explanation behavior can otherwise become difficult to understand. Table 1 summarizes the main information types used by the pipeline.

Table 1. Information used by the adaptive explanation pipeline.

Information type	Example
Scenario context	High criticality, dense traffic, urban road
Incident description	The vehicle braked because another car entered the intersection unexpectedly
Explanation constraints	Maximum three sentences, avoid jargon, stay factual
User feedback	Fewer explanations in similar low-criticality situations
System output	Explanation text, delivery mode, tone instruction, reasoning trace

The LLM is used here as a design and prototyping tool. We do not propose that unconstrained generative models should directly control communication in production vehicles. Instead, the LLM allows designers to rapidly explore how adaptive explanation strategies could behave before replacing or constraining them with more robust policies. The entire implementation uses Interfaces to make the explanation algorithms interchangeable and easy to test.

3.5 Feedback and Personalization Loop

The platform includes a feedback loop that connects user responses to future explanation behavior. After an incident, the user’s feedback is stored together with the scenario context and the explanation that was presented. This creates a structured interaction history.

In later incidents, the explanation pipeline can use this history to adapt its behavior. For example, if a user repeatedly indicates that routine lane changes do not require explanations, the system can reduce or suppress explanations in similar situations. If the user indicates that explanations should be kept in critical events, the system can continue to present explanations in those contexts.

This loop supports several forms of adaptation:

- **Frequency adaptation:** reducing or suppressing explanations in situations where the user no longer wants them.
- **Detail adaptation:** making explanations shorter or more detailed depending on feedback.

- **Tone adaptation:** adjusting how reassuring, neutral, or direct the explanation should sound.
- **Modality adaptation:** choosing whether to present an explanation through audio, text, or a less intrusive notification.

The feedback loop also raises an important design question: how permanent should a user preference be? If a user says that they do not want explanations in one type of situation, should the vehicle always suppress explanations in similar future situations? Or should it occasionally reintroduce explanations when the situation is more critical, more ambiguous, or meaningfully different? The platform is designed to make such questions testable in future studies.

3.6 Logging and Inspectability

Adaptive explanation systems need to be inspectable. If a system changes its explanation behavior over time, researchers must be able to reconstruct why a certain explanation was delivered or suppressed. The platform therefore logs scenario context, explanation output, delivery mode, user feedback, and the reasoning trace generated by the adaptive pipeline.

These logs can support debugging, comparison of explanation strategies, and later analysis of adaptive behavior. They also make it possible to inspect whether the system follows design constraints, such as avoiding invented causes or avoiding unnecessary emotional reassurance. This is particularly important when generative AI is used as part of the prototyping process.

4 Design Space for Adaptive In-Vehicle Explanations

Following dimensions and reflections outline an initial design space for context-sensitive, user-controllable, and safety-aware adaptive in-vehicle explanations.

4.1 Design Dimensions

The simulator reveals a broader design space (see Table 2) for adaptive explanations, covering timing, modality, personalization, user control, silence, and safety constraints.

Table 2. Design dimensions for adaptive in-vehicle explanations.

Dimension	Examples
Timing	Before an event, during an event, after an event, on demand
Frequency	Always explain, reduce explanations, critical-only, user-triggered
Modality	Audio, text, visual overlay, haptic cue, multimodal feedback
Personalization source	Explicit ratings, interaction history, gaze, physiology, trust estimates
Adaptation target	Content, detail, tone, modality, frequency, suppression
User control	Request explanation, dismiss explanation, change preferences, reset adaptation
Safety constraint	Factual grounding, uncertainty communication, audit logs, no speculation

4.2 Design Reflections

Developing the platform generated following design reflections for future AutomotiveUI research on adaptive in-vehicle explanations.

Adaptive Explanations Are Also About Silence: Explanation design concerns not only what to communicate, but also when not to communicate. In automated vehicles, silence may indicate a routine situation, repeated exposure to similar behavior, or intentional suppression based on user preferences, making it part of the interface. However, users may interpret silence as either a sign that everything is normal or a failure to explain something important. Systems may therefore need to communicate their explanation policy, for example through a minimal notification that an explanation was suppressed based on a configured preference. Because such notifications can themselves become unnecessary, the platform enables researchers to explore this tension.

Scenario Labels Are Not Enough: The platform represents situations through labels such as criticality, traffic density, and road type. Although useful for scenario authoring and experimental control, these labels cannot fully capture explanation needs. Two high-criticality urban events, for example, may differ because one is visually obvious while the other is difficult to interpret; similarly, the need to explain a lane change may depend on whether its cause is visible. Adaptive explanations therefore require richer situational features, including cause visibility, action predictability, the involvement of vulnerable road users, maneuver suddenness, and user attention. The simulator offers a controlled setting for determining which features matter most.

Personalization Must Be Balanced With Safety: Personalization can improve explanations by reducing unnecessary information and accommodating individual preferences, but preferences alone should not determine explanation behavior. Repeated requests for fewer explanations could suppress information needed for safety, accountability, or trust calibration. Systems may therefore need to distinguish optional explanations from those that should remain available in safety-relevant situations and apply override rules in critical or uncertain conditions.

LLMs Are Useful for Prototyping, but Need Constraints: The LLM-based pipeline supports rapid prototyping of explanation styles and adaptive policies, which is valuable during early design. However, generative models may produce incorrect, overconfident, or inconsistent outputs in a safety-critical domain. The platform therefore uses the LLM as a prototyping tool rather than a production-ready component, constraining it to provided scenario information, prohibiting unsupported speculation, and requiring structured, inspectable outputs. Insights from the prototype can inform more constrained approaches, including template-based explanations, rule-based policies, and hybrid architectures combining symbolic safety rules with generative language.

User Control Should Be Part of Adaptation: Although adaptive systems often infer preferences automatically, vehicle users may want to understand and control such adaptation. When explanations are suppressed, users should be able to determine why and revise the relevant preference; otherwise, adaptation may become opaque. The platform therefore makes feedback explicit, allowing users to keep, reduce, or omit explanations in similar situations. Future versions could add preference menus, reset options, and on-demand explanations, enabling comparisons between automatic and user-configurable adaptation.

5 Planned Future Work

Planned extensions include expanding the scenario library with more varied road users, environmental conditions, weather, visibility, and event types; extending the explanation interface with visual overlays, ambient cues, and on-demand controls; and incorporating implicit signals such as gaze, head orientation, interaction behavior, and physiological measures. Future empirical studies will enable controlled comparisons of proactive, on-demand, and

hybrid adaptation strategies while strengthening safety and validation mechanisms related to scenario grounding, uncertainty communication, and the prevention of explanation suppression in safety-relevant situations.

6 Conclusion

The paper contributes a research instrument and design framing for studying explanation timing, suppression, personalization, user control, modality, and safety constraints in LLM-supported automotive interfaces. Researchers can use the platform to create controlled driving scenarios, vary when and how explanations are presented, compare proactive, on-demand, and hybrid adaptation strategies, collect user feedback across repeated interactions, and inspect why explanations were generated or suppressed. They can also extend it with new traffic situations, explanation modalities, personalization signals, and safety mechanisms, while the lessons from its development help outline an initial design space for adaptive in-vehicle explanations.

References

- [1] Srivatsan Chakravarthi Kumaran, Toam Bechor, and Hadas Erel. 2024. A Social Approach for Autonomous Vehicles: A Robotic Object to Enhance Passengers' Sense of Safety and Trust. In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction (HRI '24)*. Association for Computing Machinery, New York, NY, USA, 86–95. doi:10.1145/3610977.3634998
- [2] Mark Colley, Christian Bräuner, Mirjam Lanzer, Marcel Walch, Martin Baumann, and Enrico Rukzio. 2020. Effect of Visualization of Pedestrian Intention Recognition on Trust and Cognitive Load. In *12th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI '20)*. Association for Computing Machinery, New York, NY, USA, 181–191. doi:10.1145/3409120.3410648
- [3] Mark Colley, Oliver Speidel, Jan Strohbeck, Jan Ole Rixen, Jan Henry Belz, and Enrico Rukzio. 2024. Effects of Uncertain Trajectory Prediction Visualization in Highly Automated Vehicles on Trust, Situation Awareness, and Cognitive Load. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 7, 4 (Jan. 2024), 153:1–153:23. doi:10.1145/3631408
- [4] Rudresh Dwivedi, Devam Dave, Het Naik, Smiti Singhal, Rana Omer, Pankesh Patel, Bin Qian, Zhenyu Wen, Tejal Shah, Graham Morgan, and Rajiv Ranjan. 2023. Explainable AI (XAI): Core Ideas, Techniques, and Solutions. *ACM Comput. Surv.* 55, 9 (Jan. 2023), 194:1–194:33. doi:10.1145/3561048
- [5] Yannick Forster, Frederik Naujoks, and Alexandra Neukum. 2017. Increasing anthropomorphism and trust in automated driving functions by adding speech output. In *2017 IEEE Intelligent Vehicles Symposium (IV)*. IEEE Press, Los Angeles, CA, USA, 365–372. doi:10.1109/IVS.2017.7995746
- [6] Gwangbin Kim, Dohyeon Yeo, Taewoo Jo, Daniela Rus, and SeungJun Kim. 2023. What and When to Explain? On-road Evaluation of Explanations in Highly Automated Vehicles. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 7, 3 (Sept. 2023), 104:1–104:26. doi:10.1145/3610886
- [7] Soyeon Kim, René van Egmond, and Riender Happee. 2024. How manoeuvre information via auditory (spatial and beep) and visual UI can enhance trust and acceptance in automated driving. *Transportation Research Part F: Traffic Psychology and Behaviour* 100 (Jan. 2024), 22–36. doi:10.1016/j.trf.2023.11.007
- [8] Jeamin Koo, Jungsuk Kwac, Wendy Ju, Martin Steinert, Larry Leifer, and Clifford Nass. 2015. Why did my car just do that? Explaining semi-autonomous driving actions to improve driver understanding, trust, and performance. *International Journal on Interactive Design and Manufacturing (IJIDeM)* 9, 4 (Nov. 2015), 269–275. doi:10.1007/s12008-014-0227-2
- [9] Anton Kuznetsov, Balint Gyevnar, Cheng Wang, Steven Peters, and Stefano V. Albrecht. 2024. Explainable AI for Safe and Trustworthy Autonomous Driving: A Systematic Review. *IEEE Transactions on Intelligent Transportation Systems* 25, 12 (Dec. 2024), 19342–19364. doi:10.1109/TITS.2024.3474469
- [10] Daniel Omeiza, Helena Webb, Marina Jirotko, and Lars Kunze. 2021. Explanations in autonomous driving: A survey. *IEEE Transactions on Intelligent Transportation Systems* 23, 8 (2021), 10142–10162.
- [11] Mersedeh Sadeghi, Daniel Pöttgen, Patrick Ebel, and Andreas Vogelsang. 2024. Explaining the Unexplainable: The Impact of Misleading Explanations on Trust in Unreliable Predictions for Hardly Assessable Tasks. In *Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization (UMAP '24)*. Association for Computing Machinery, New York, NY, USA, 36–46. doi:10.1145/3627043.3659573
- [12] Philipp Wintersberger, Anna-Katharina Frison, Andreas Riener, and Tamara von Sawitzky. 2018. Fostering user acceptance and trust in fully automated vehicles: Evaluating the potential of augmented reality. *PRESENCE: Virtual and Augmented Reality* 27, 1 (2018), 46–62.
- [13] Philipp Wintersberger, Hannah Nicklas, Thomas Martlbauer, Stephan Hammer, and Andreas Riener. 2020. Explainable Automation: Personalized and Adaptive UIs to Foster Trust and Understanding of Driving Automation Systems. In *12th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI '20)*. Association for Computing Machinery, New York, NY, USA, 252–261. doi:10.1145/3409120.3410659

Received 18 June 2026

Manuscript submitted to ACM